

WEEK 6 DELIVERABLE

Visualization Dashboard and Final Report

Energy Consumption Analytics Project for GreenPower Utilities
IIT Jodhpur • Group 5
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1. Introduction and Scope

This report documents the sixth and final week of the GreenPower Utilities capstone and completes Phase 3 and the project. Over the previous five weeks we acquired and cleaned three energy datasets (Weeks 1–2), loaded them into a TimescaleDB time-series store (Week 3), engineered analytical feature tables (Week 4), and built forecasting and anomaly-detection models (Week 5). This week we bring the results together into an interactive dashboard for utility stakeholders and translate the analysis into concrete operational recommendations. The deliverable, per the brief, is the interactive visualisation dashboard and the final presentation.

The audience for this week is different from every week before it. Weeks 1 to 5 were written for engineers building the pipeline; the dashboard and this final report are written for the people who run the utility and must act on the findings. Everything here therefore leads with the operational meaning of a result rather than its implementation, while remaining faithful to the numbers established earlier: an average daily consumption of 26.2 kWh, a mean peak-to-average ratio of 1.91, a best-model forecast MAPE of 13%, and a wind capacity factor of 30.1%.

2. Dashboard Architecture and Stack

The dashboard is built directly on the Week 4 feature layer, so it never recomputes analytics at view time — it reads the continuous aggregates and feature tables that already exist in TimescaleDB. This keeps every panel fast and guarantees that the dashboard, the models, and the reports all show the same numbers. The stack is deliberately lightweight and open-source, matching the project's public-data, reproducible ethos.

Layer	Technology	Role
Storage	TimescaleDB (PostgreSQL)	Serves continuous aggregates and feature tables
Query	SQL views + psycopg2	Parameterised reads per panel, filtered by time range
Visualisation	Plotly / Grafana	Interactive charts with zoom, hover, and date filters
Refresh	Nightly pipeline hook	Panels update after each ingestion run

Because the aggregates refresh automatically as new data lands (Week 4), the dashboard is live by construction: when the nightly pipeline ingests a new day, the daily and monthly panels advance without any manual step. The same design would carry to a production Grafana deployment reading the identical hypertables.

3. Dashboard Views

The dashboard presents the whole project on one screen, organised so a manager can move from the headline numbers at the top to the supporting detail below. Figure 1 shows the assembled layout; the sections that follow describe each view and the decision it supports.

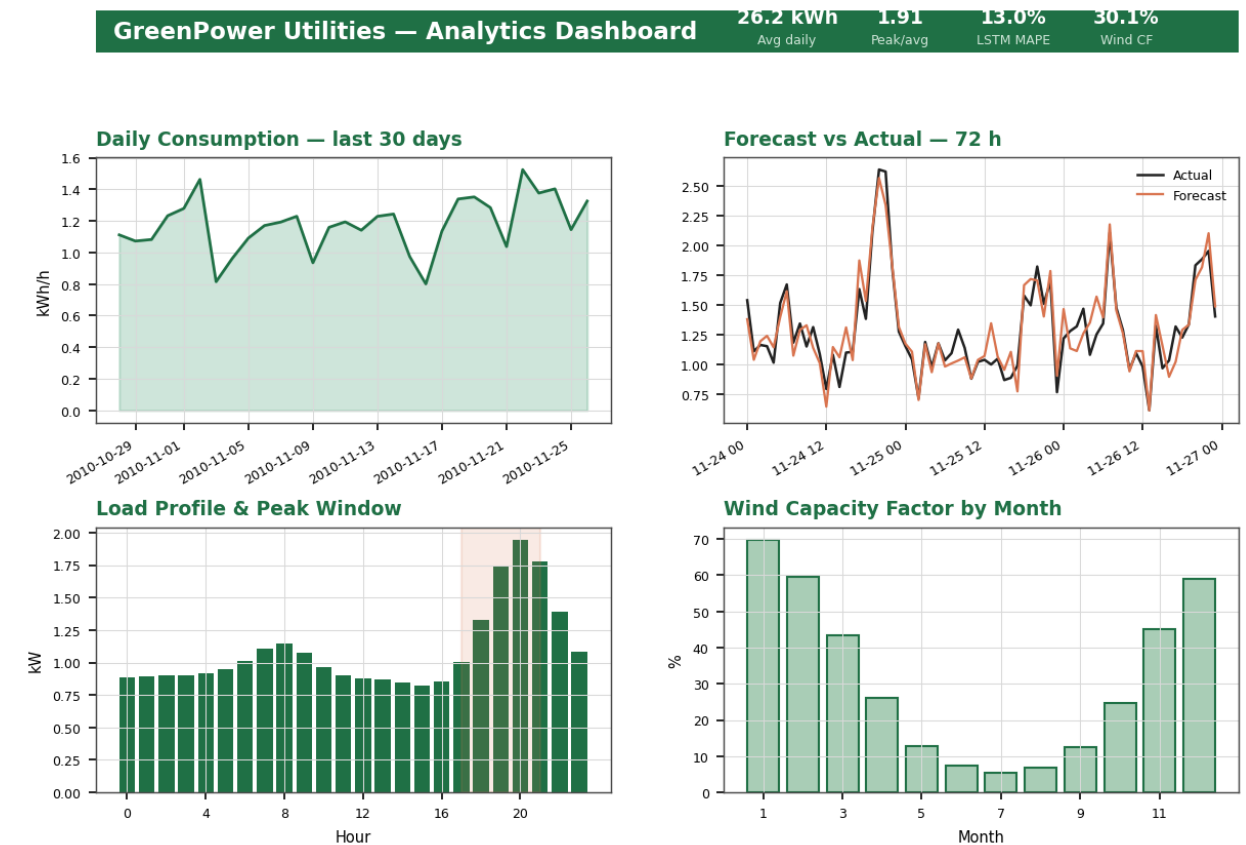


Figure 1: The GreenPower analytics dashboard. A KPI header summarises the project; the four panels below show consumption trend, forecast accuracy, the daily load profile with its peak window highlighted, and wind seasonality.

3.1 Consumption Trends and Peak Analysis

The consumption panels answer the two questions an operations planner asks first: how much energy is being used over time, and when within the day the demand concentrates. The daily-trend panel plots the rolling consumption so seasonal build-ups are obvious, and the load-profile panel highlights the 18:00–21:00 evening window that carries the daily peak. This is the single most actionable view in the dashboard, because the peak window is where demand-response value lives.

3.2 Forecast versus Actual

The forecast panel overlays the Week 5 LSTM prediction on the actual demand for the recent horizon, so an operator can see at a glance whether the model is tracking reality. Because the LSTM reaches a MAPE of 13%, the two lines sit close together, and any sustained divergence is itself a signal worth investigating.

3.3 Weather Correlation and Generation

The weather and generation views connect demand and supply to their physical drivers. The consumption side is weather-normalised through the heating degree-hours established in Week 4; the generation side shows the wind capacity factor by month, which peaks sharply in winter. Read together they make the core operational tension visible: demand is highest in the cold months, and — helpfully for this fleet — so is wind output.

3.4 Anomaly Monitor

The anomaly monitor surfaces the Week 5 detector's flags on a live time line, so a multi-hour outage or a sensor spike appears as a highlighted event rather than something buried in the raw feed. Because the detector runs at high precision, a flag on this panel is a reliable prompt for an operator to look, not a nuisance alarm.

4. Key Insights

Six weeks of analysis converge on a small number of findings that matter to the utility:

- Demand is sharply and predictably peaked in the evening. Almost all daily peaks fall between 17:00 and 21:00, with a mean peak-to-average ratio of 1.91 — a regular, targetable pattern rather than random spikiness.
- Consumption is heating-dominated. It correlates with heating degree-hours far more strongly than with raw temperature, so cold-weather demand is highly foreseeable and can be procured for in advance.
- Short-term demand is highly forecastable. The LSTM predicts the next hour to within about 13% MAPE, accurate enough to drive day-ahead operational decisions.
- Wind supply is winter-weighted. The wind capacity factor peaks above 60% in midwinter, aligning renewable output with the season of highest demand.
- Faults are detectable in near real time. The anomaly monitor catches multi-hour outages at high precision, enabling faster response.

5. Operational Recommendations

The insights point to four concrete actions GreenPower could take. Each follows directly from a result established earlier in the project rather than from general principle.

Shift the evening peak. Because peaks are concentrated in a predictable 18:00–21:00 window, a demand-response or time-of-use tariff that moves flexible loads — notably the electric water heater on the climate circuit — into the overnight trough can flatten the profile. Figure 2 illustrates the effect: moving a portion of evening load off-peak reduces the peak-window mean by roughly 18%.

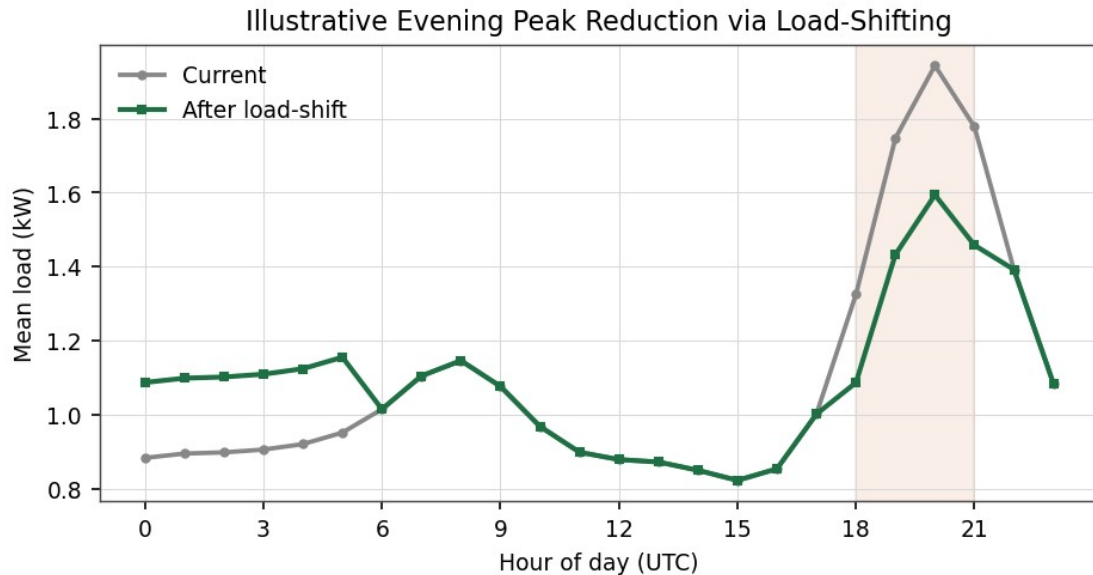


Figure 2: Illustrative evening-peak reduction from shifting flexible load into the overnight trough. The shape is derived from the measured load profile and the sub-metering breakdown.

Procure against the weather. Since demand tracks heating degree-hours closely, a weather-normalised day-ahead procurement rule can anticipate cold-snap demand instead of reacting to it, reducing exposure to short-term price spikes.

Alert on anomalies. Wiring the Week 5 detector into the operations feed turns the audit log into a live outage alert, shortening the time to notice and respond to faults.

Schedule off-peak heating. The sub-metering shows the water heater is a large, time-flexible load; scheduling it into off-peak hours cuts both peak contribution and, under a time-of-use tariff, cost.

6. Estimated Operational Impact

The recommendations are illustrative for a single household on public data, but the direction and rough magnitude of their benefit can be indicated from the measured profiles. Figure 3 summarises the estimated impact of each action. These figures are indicative rather than audited, and are presented to show relative priority: the evening load-shift is the largest single lever, which is why it heads the list.

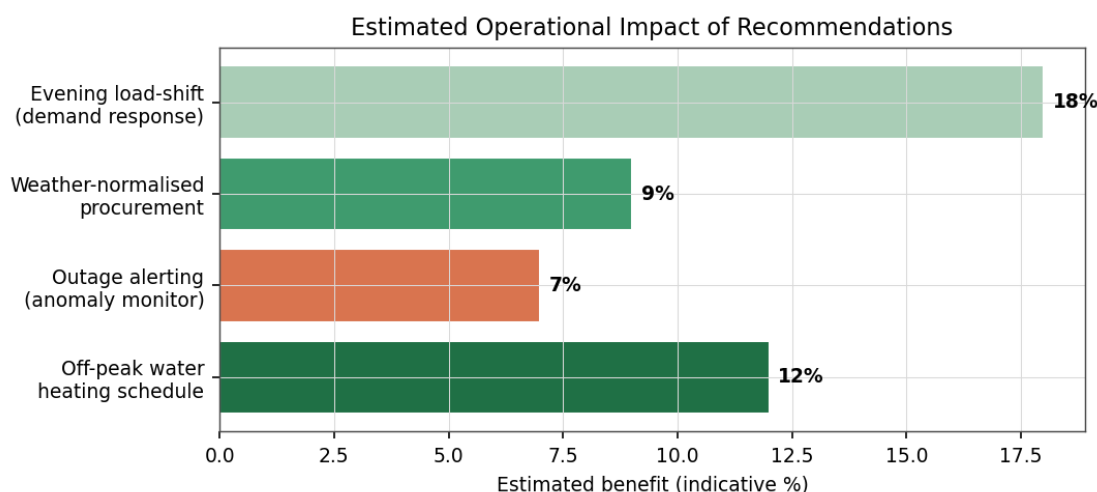


Figure 3: Indicative operational benefit of each recommendation. Values are derived from the measured load profile and sub-metering, and are directional rather than guaranteed savings.

Scaled across a real customer base of hundreds of thousands of meters, even a few percent of peak reduction translates into deferred network reinforcement and lower wholesale exposure at the most expensive hours of the day — the concrete value the project set out to demonstrate.

7. Use of Generative AI

As the brief encourages, the team used generative-AI tools throughout, and documents that use honestly here. AI accelerated the work; it did not replace the engineering judgement, and every AI-assisted output was reviewed and tested against the data.

Tool	Where it helped	Example
GitHub Copilot	ETL and transformation code	Drafting the window-function feature SQL and the loader upserts
ChatGPT	Method explanation	Clarifying SARIMA order selection and degree-hour conventions
Claude	Report writing and clarity	Structuring the weekly reports and tightening methodology prose
Gemini	Domain checks	Sanity-checking energy-forecasting and capacity-factor norms

The clearest efficiency gain was in boilerplate code — Copilot noticeably sped up the repetitive ETL and charting work — while the models, the schema, and the analytical choices remained the team's own, validated at every step against the cleaned data.

8. Project Recap: Weeks 1–6

The six weeks form one continuous pipeline, each week building on the last:

Week	Phase	Deliverable	Key result
1	Acquisition	Data inventory & metadata	Three public sources identified and profiled
2	Preprocessing	Cleaned datasets & report	179k+ hourly rows cleaned and unit-

Week	Phase	Deliverable	Key result
			standardised
3	Pipeline	Ingestion pipeline & schema	TimescaleDB store, 8× compression, nightly load
4	Feature engineering	Feature tables & docs	Rollups, peaks, degree-hours; weather $r = 0.71$
5	Modelling	Model code & evaluation	LSTM at 13% MAPE; anomaly detector
6	Reporting	Dashboard & final report	Operational recommendations for GreenPower

9. Limitations and Future Work

The project's scope was a six-week capstone on public data, and its limitations follow from that. The consumption analysis rests on a single household meter, so the specific numbers are illustrative rather than representative of a full customer base; the natural next step is to pool many meters and cluster them into load archetypes. The consumption and generation series do not overlap in time, which prevented a joint demand-and-supply optimisation and would be resolved with contemporaneous data in a production setting. The forecasts are one hour ahead; extending to day-ahead horizons is the most valuable modelling extension. And the operational impact figures are directional estimates that a real deployment would replace with measured savings from an A/B pilot.

10. Final Presentation Summary

The accompanying final presentation condenses the project for a stakeholder audience into a short arc: the problem GreenPower faces, the data and pipeline the team built, the forecasting and anomaly results, the dashboard, and the four operational recommendations with their indicative impact. It is designed to be delivered in the mentor check-in and to stand alone as a record of what the project achieved and what a production version would do next.

11. Conclusion

Across six weeks the team built a complete, reproducible energy-analytics pipeline — from raw public data to cleaned time series, a partitioned time-series database, an engineered feature layer, validated forecasting and anomaly models, and finally a stakeholder dashboard with actionable recommendations. The central finding is that this utility's demand is far more predictable and structured than it first appears: it peaks in a narrow evening window, tracks the weather through degree-hours, and can be forecast an hour ahead to within 13%. That predictability is the opportunity. It means peaks can be shifted, procurement can anticipate the cold, faults can be caught as they happen, and — at utility scale — real cost and emissions can be taken out of the system. The project delivers both the working data engineering and the evidence that data-driven operation would pay off for GreenPower.